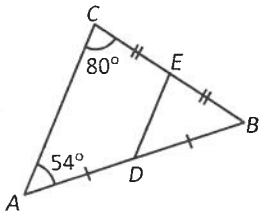
There are 20 questions in Section A and 10 questions in Section B.

The diagrams in this paper are not necessarily drawn to scale.

Choose the best answer for each question.

Section A (1-20)

1. In the figure, D and E are the mid-points of AB and CB respectively. Find $\angle BED$.



A. 90°
B. 46°
C. 54°
D. 80°

2. $4x^2 - 25xy + 39y^2 =$
   A. $(x - 13y)(4x - 3y)$.
   B. $(x + 13y)(4x - 3y)$.
   C. $(x - 3y)(4x - 13y)$.
   D. $(x + 3y)(4x - 13y)$.

3. Which of the following has $x - 2$ as a factor?
   A. $x^2 + 3x + 10$
   B. $x^2 + x - 6$
   C. $x^2 - 2x - 8$
   D. $2x^2 + 7x - 4$

4. $y^3(y + y^0) =$
   A. y^4.
   B. $2y^4$.
   C. $y^4 + 1$.
   D. $y^4 + y^3$.

5. Calculate $1001_{10} - 1001_2$
   A. 0
   B. 992
   C. 8
   D. 993

6. $(3^y)(5^y) =$
   A. 15^{2y}.
   B. 8^y.
   C. 8^{2y}.
   D. 15^y.

7. A man gained weight by 5% in July and lost weight by 3% in August. What is his percentage change in weight in these 2 months?
   A. +2.0%
   B. +1.85%
   C. –2.0%
   D. –1.85%

8. When a number increases by 25%, it becomes 80. The original number is
   A. 60.
   B. 100.
   C. 64.
   D. 107

9. If the angle of sector and the radius of a sector are increased by x% and 20% respectively, its area is increased by 100%. Then $x =$
   A. 29.
   B. 39.
   C. 49.
   D. 59.

10. Winnie deposits $4000 in a bank at an interest rate of 6% per annum. Find the simple interest that Winnie will receive after 3 months.
   A. $40
   B. $720
   C. $60
   D. $240

11. A sum of $30 000 is deposited at an interest rate of 3% per annum for 2 years, compounded yearly. Find the amount correct to the nearest dollar.

A. $30 900
B. $31 800
C. $31 827
D. $33 708

12. The price of a phone depreciates at a rate of 15% yearly. The price of a phone is now $2500. What was its price 2 years ago?

A. $3570
B. $3460
C. $1806
D. $1750

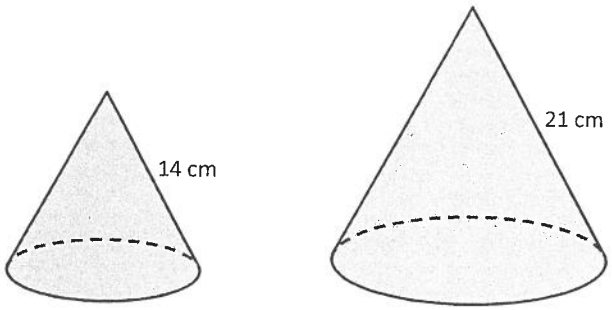
13. *Table for salary tax is shown below:*

	Net chargeable income	Rate
On the First	$50 000	2%
On the Next	$50 000	6%
On the Next	$50 000	10%
On the Next	$50 000	14%
Remainder		17%

If Mr. Wong paid $9 400 as salaries tax, find his net chargeable income.

A. $67 000
B. $110 000
C. $153 000
D. $200 000

14. In the figure, the lengths of the corresponding slant heights of two similar cones are 14 cm and 21 cm respectively.



If the volume of the smaller cone is V cm^3, then the volume of the larger cone is

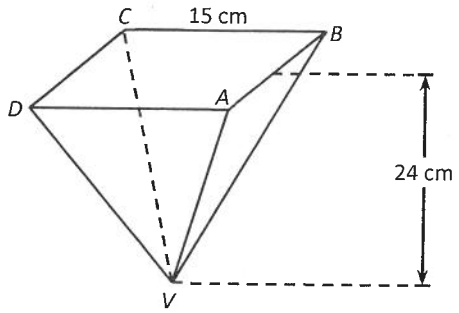
A. $1.5V$ cm^3.

B. $2.25V$ cm^3.

C. $3.375V$ cm^3.

D. $8\ V$ cm^3.

15. The figure shows an inverted pyramid $VABCD$ of height 24 cm. Its base is a square of side 15 cm.



The volume of the pyramid is

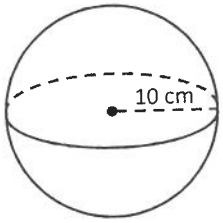
A. 1440 cm^3.

B. 1800 cm^3.

C. 5400 cm^3.

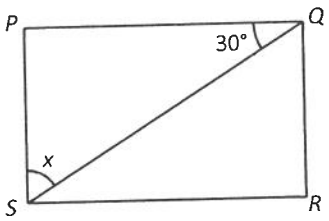
D. 5650 cm^3.

16. The figure shows a sphere of radius 10 cm. Find the volume of the sphere, correct to the nearest cm^3.



A. 1256 cm^3
B. 1333 cm^3
C. 4189 cm^3
D. 33510 cm^3

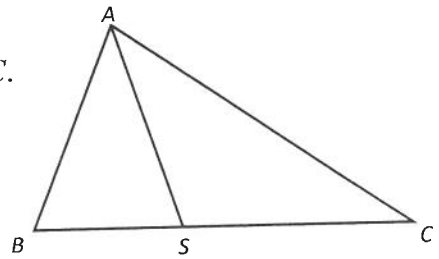
17. In the figure, $PQRS$ is a rectangle. Find x.



A. 30°
B. 45°
C. 60°
D. 70°

18. In $\triangle ABC$ as shown, S is a point on BC. $\angle BAS = \angle SAC$. AS **MUST**

A. bisect BC.
B. be a perpendicular bisector of $\triangle ABC$.
C. be perpendicular to BC.
D. be an angle bisector of $\triangle ABC$.



19. The mini-figure of Yao is shown.

The mini-figure of Yao and actual Yao are similar figures.



The height of the mini-figure is 15 cm and the height of actual Yao is 2.29 m. Find the ratio of the total surface area of the mini-figure to the total surface area of actual Yao.

A. 15 : 2.29

B. 15 : 229

C. $15^2 : 229^2$

D. $15^3 : 229^3$

20. Which of the following cannot be factorized?

I $-49x^2 + 25y^2$

II $3 - 8x^3$

III $x^2 + 4x + 100$

A. I & II only

B. I & III only

C. II & III only

D. I , II & III

Section B (21-30)

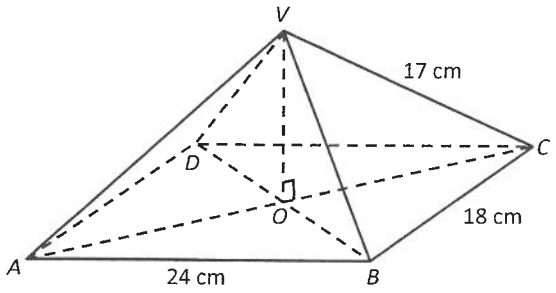
21. $5s(s-t)-2t(9s+5t)=$
    A. $(s-5t)(5s+2t)$.
    B. $(s+5t)(5s-2t)$.
    C. $(s-10t)(5s+t)$.
    D. $(s+10t)(5s-t)$.

22. $4s^2-7st+3t^2-s+t=$
    A. $(s+t)(4s+3t-1)$.
    B. $(s-t)(4s+3t+1)$.
    C. $(s+t)(4s-3t-1)$.
    D. $(s-t)(4s-3t-1)$.

23. Arrange the following number in descending order.
    I 10000000100_2
    II 1026_{10}
    III 10000000011_2
    A. I, III, II
    B. III, I, II
    C. I, II, III
    D. III, II, I

24. A sum of money is deposited at an interest rate of 10% per annum for n years, compounded yearly. Find the minimum time for this amount to be doubled.
    A. 6 years
    B. 7 years
    C. 8 years
    D. 9 years

25. The figure shows a right pyramid $VABCD$ of slant edge 17 cm. Its base is a rectangle of length 24 cm and width 18 cm. The diagonals AC and BD intersect at O.



The volume of the pyramid is

A. 1152 cm^3.

B. 1830 cm^3.

C. 2160 cm^3.

D. 2448 cm^3.

26. If the diameter of a sphere is increased by 100%, then the total surface area of the sphere is increased by

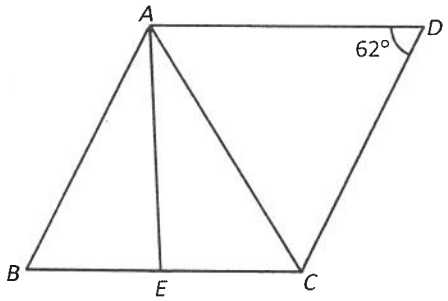
A. 100%.

B. 125%.

C. 300%.

D. 400%.

27. In the figure, $ABCD$ is a rhombus. E is a point on BC such that $\angle BAE = \angle CAE$. $\angle ADC = 62°$. Find $\angle BEA$.



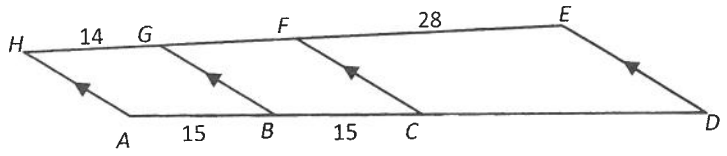
A. 87.5°

B. 88°

C. 88.5°

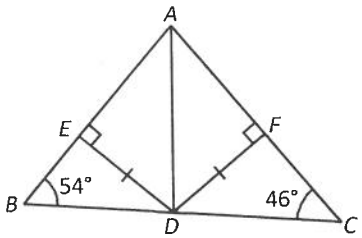
D. 90°

28. In the figure, $ABCD$ and $HGFE$ are straight lines. Find the length of CD.



A. 24
B. 26
C. 28
D. 30

29. In the figure, AEB, AFC and BDC are straight lines. $DE = DF$, $\angle AED = \angle AFD = 90°, \angle ABD = 54°$ and $\angle ACD = 46°$. Find $\angle ADC$.



A. 90°
B. 92°
C. 94°
D. 96°

30. $y^{k+7} \cdot y^{7-k} =$
A. y^{14}.
B. y.
C. y^2.
D. y^{2k}.

END OF PAPER